Name: **V#:**

Sample Mid-term Exam

CMSC 440, Spring 2023

Data Communications and Networking

Virginia Commonwealth University

Directions:

Question	Points	Score
I (2Px10Q)	20	
II.1	5	
II.2	9	
II.3	12	
II.4	4	
II.5	6	
II.6	10	
II.7	12	
II.8	10	
II.9	12	
Total	**100**	

1. This ***Instructions*** page is not part of the exam. ***Take your time reading these instructions carefully***. Do not flip it and start the exam until you are instructed to do so.
2. This is a closed book, closed notes, closed internet, closed neighbor, pencil and paper exam.
3. This exam represents ***25% of the total CMSC440 grade.***
4. This exam is designed to be completed in **75 minutes**. Turn immediately the exam when we call for it.
5. The exam has two types of questions: ***10 questions*** of ***Fill in the Blank*** with a total of *20* ***points*** and ***10 questions*** of ***Short Answer*** with a total of ***80 points.***
6. Read each question carefully, paying attention to detail, including the units given in the problem statement and the units asked for in the question.
7. Write answers in the spaces provided. Indicate your final answer clearly. Cleanly erase or cross out any work you do not want graded.
8. Show your work and explain your reasoning where appropriate; this will help to earn partial credit.
9. If you need additional space, use the back of the page.
10. Electronic devices except calculators must be completely turned off and stowed in your backpack.
11. Write your name and V# on each sheet of paper

<u>Good Luck!</u>

Name: **V#:**

Part I: Terms [20 points] – Fill in the blank with the appropriate term. [2 points each]

Circuit Switching Network, Packet Switching Network, Bandwidth/Throughput, Socket, Traceroute, Multiplexing, 3-Way Handshake, Proxy, DNS, Protocol, Router, Stop-and-Wait, Stop-and-Forward, Forwarding, Routing, Port, non-persistent HTTP, Multiplexing, Propagation delay, Parallelism, Pipelining, Transmission delay, Persistent HTTP, 2-Way Handshake, Access link.

1) A ____ defines the format, order of messages sent and received among network entities, and actions taken on message transmission and receipt.

2) ____ is a type of network in which a physical path is obtained for and dedicated to a single connection between two end-points in the network for the duration of the connection.

3) A ___________________________________ is defined as "the endpoint" in a connection for sending and receiving data between two programs running on the network. It is created and used with a set of programming requests or "function calls".

4) The purpose of ____ is to uniquely identify different applications or processes running on a single computer and thereby enable them to share a single physical connection to a packet-switched network like the Internet.

5) The _________________________________ is a hierarchical distributed naming system for computers, services, or any resource connected to the Internet or a private network.

6) ____ is the rate (bits/time unit) at which bits are transferred between a sender and receiver.

8) A ____ is a machine that is used to satisfy client HTTP requests without contacting the origin server. Nearby clients send all of their HTTP requests to this machine.

9) ____ is a computer network diagnostic tool for displaying the route (path) and measuring transit delays of packets across an Internet Protocol (IP) network.

10) In ___________________________, the sender is allowed to send multiple packets without waiting for an acknowledgment.

Part II: Short Answer [80 points]

1. [5 points] Suppose that all of the network sources send data at a constant bit rate. Would packet switching or circuit-switching be more desirable in this case? Why?

2. [9 points]

a. [5 points] List the 5 Internet protocol layers in order from top to bottom.

1.
2.
3.
4.
5.

b. [2 points] An end-system processes up to which layer?

c. [2 points] A router processes up to which layer?

3. [12 points] Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (this is the actual content of an HTTP GET message). The characters **<cr><lf>** are carriage return and line feed characters (that is, the italicized character string **<cr>** in the text below represents the single carriage-return character that was contained at that point in the HTTP header). Answer the following questions.

GET /cs453/index.html HTTP/1.1<cr><lf>Host: gaia.cs.umass.edu**<cr><lf>**User-Agent: Mozilla/5.0 (Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax) **<cr><lf>**Accept:ext/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,image/png,*/*;q=0.5**<cr><lf>**Accept-Language: en-us,en;q=0.5..Accept-Encoding: zip,deflate**<cr><lf>**Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7..Keep-Alive: 300**<cr><lf>**Connection:keep-alive**<cr><lf><cr><lf>**

a. [3 point] What is the URL of the document requested by the browser? Make sure you give the hostname and the file name parts of the URL.

b. [3 point] Is a Netscape or an Internet Explorer browser making the request?

c. [3 point] Is the browser requesting a non-persistent or a persistent connection?

d. [3 points] What is the IP address of the computer on which the browser is running?

4. [4 points] Describe how packet loss at a router typically occurs.

5. [6 points] In Go-back-N protocol:

a. [3 points] Does this protocol have a timer for each unacknowledged packet?

b. [3 points] When a timer expires, what happens?

6. [10 points] Consider the following:

a. [5 points] Suppose a process in host C has a UDP socket with port number 787. Suppose host A and host B each send a UDP segment to host C with destination port number 787. Will both of these segments be directed to the same socket at host C? If so, how will the process at host C know that these segments originated from two different hosts?

b. [5 points] Suppose that a Web server runs in host C on port 80. Suppose this Web server uses persistent connections, and is currently receiving requests from two different hosts: A and B. Are all of the requests being sent through the same socket at host C? If they are being passed through different sockets, do both of the sockets have port 80?

7. [12 points] Consider sending a packet of F bits over a path of Q links. Each link transmits at R bps. The network is lightly loaded so that there are no queuing delays. Propagation delay is also negligible.

a. [4 points] Suppose the network is a packet-switched datagram network and a connection-oriented service is used. Suppose each packet has h x F bits of header where 0<h<1. Assuming t_s setup time, how long does it take to send the packet?

b. [4 points] Suppose that the network is a circuit-switched network. Furthermore, suppose that the transmission rate of the circuit between source and destination is R/24 bps. Assuming t_s setup time and no bits of header appended to the packet, how long does it take to send the packet?

c. [4 points] When is the delay longer for packet switching than for circuit switching assuming h=0.5?

8. [10 points] Consider the following network configuration for a small college campus. The campus is connected to the Internet via a 2 Mbps access link. The only traffic on the local campus network is generated by web browsers using HTTP to access web objects stored by origin servers on the public Internet. Measurements on this network indicate that the campus browsers in aggregate send 35 HTTP requests per second, and so you can assume HTTP responses (including the browsing objects) arrive at the access link at a rate of 35 HTTP response per second. The mean size of all HTTP responses is 40,000 bits.

- **a) [5 points]** What is the inbound (incoming) traffic intensity on the 2 Mbps access link without a proxy cache?

- **b) [5 points]** A web proxy cache has been installed on the campus network and all the browsers have been configured to send all HTTP requests to the proxy server. The proxy server has a hit rate of 45%. What is the inbound traffic intensity on the access link?

10. [12 points] A simple synchronized message exchange protocol. Consider two network entities: A and B, which are connected by a perfect bi-directional channel (that is, any message sent will be received correctly; the channel will not corrupt, lose, or re-order packets). A and B are to deliver data messages to each other in an alternating manner: first A must deliver a message to B, then B must deliver a message to A, then A must deliver a message to B, and so on.

Draw a FSM specification for this protocol (one FSM for A and one FSM for B). Don't worry about a reliability mechanism here; the main point is to create a FSM specification that reflects the synchronized behavior of the two entities. You should use the following events and actions, which have the same meaning as protocol rdt1.0: **rdt_send(data),**
packet = make_pkt(data), udt_send(packet), rdt_rcv(packet), extract (packet,data), deliver_data(data).

Make sure that your protocol reflects the strict alternation of sending between A and B. Also, be sure to indicate the initial states for A and B in your FSM description.

Scratch page